

AEM-X100 — Preventive Maintenance Guide

Equipment: AEM-X100 Wafer Handling System

Document ID: SYN-X100-PREVENTIVE_MAINTENANCE **Revision:** A **Status:** Synthetic Training Data

IMPORTANT: This document is entirely fictional and created for the AAS Practice Module prototype. It is not an AEM document, does not represent an actual AEM product or maintenance procedure, and must not be used for real equipment servicing.

Purpose: Provide synthetic maintenance intervals and inspection topics for knowledge retrieval and safe escalation testing.

1. Maintenance Principles

- Preventive maintenance activities must follow the approved site maintenance plan.
- The fictional intervals below are intentionally simplified for the prototype and are not real equipment requirements.
- Any maintenance task requiring access to a protected area must follow Safety Procedure before work begins.

2. Synthetic Maintenance Schedule

Interval	Activity	Reference
Daily	Visual condition check; review alarms	Equipment Overview / Error Code Reference
Weekly	Inspect accessible sensor surfaces and transfer area	Sensor Troubleshooting
Monthly	Review transfer-position behaviour and maintenance counters	Sensor Troubleshooting
Quarterly	Review safety-interlock status and maintenance records	Safety Procedure
As required	Investigate repeated alarms or abnormal trends	Escalation Procedure

3. Maintenance Record Requirements

- Record equipment identifier, date, activity, observed condition, alarm codes and outcome.
- Do not mark a maintenance activity complete when the required evidence is missing.
- If a required inspection cannot be completed safely, record the condition and escalate.

4. AI Assistant Behaviour

- The assistant may summarize documented maintenance requirements.
- The assistant must not invent maintenance intervals or claim that a task was completed.
- For physical maintenance instructions, the assistant should cite the source document and warn when the retrieved evidence is incomplete.

End of synthetic document.